

APEX STRATA
CONSULTING

Company Profile

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Company Overview

Apex Strata Consulting is a specialized geotechnical engineering consultancy dedicated to serving the mining industry. Structured as a lean, low-overhead enterprise, we leverage a highly agile model to deliver top-tier geotechnical services without the burdensome corporate costs associated with larger multinational firms. We exist to close a critical gap in the market, providing small to medium-scale mining operations with access to world-class geotechnical expertise. By translating complex rock mechanics into practical, field-oriented tools, we help our clients mitigate severe operational risks, prevent ground failures, and optimize resource recovery.

Mission and Objectives

Our Mission: To empower small- and medium-scale mines with agile, affordable, and actionable rock engineering solutions, transforming safety and productivity from a high-cost luxury into an accessible necessity.

Our Objective: We strive to deliver services that meet our clients' unique needs and add value to their projects. Whether it is a baseline hazard assessment, an intricate underground support design, slope stability analysis, or a comprehensive Tailings Storage Facility (TSF) stability review, our goal is to build long-term, sustainable safety cultures and operational stability for every client we partner with.

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Values

- **Integrity:** We hold ourselves to the highest ethical and engineering standards, delivering honest, data-driven assessments and transparent pricing.
- **Customer Focus:** We do not just deliver reports; we deliver solutions. We prioritize clarity, ensuring our recommendations are highly actionable for on-site personnel.
- **Excellence:** We apply rigorous engineering analysis, advanced numerical modeling, and industry best practices to guarantee the safety and structural integrity of every project.

Service Offerings

Our services are designed as a logical progression, allowing clients to engage at a level that matches their operational tempo and budget:

- **Geotechnical Investigations & Core Logging:** Comprehensive site investigations, soil profiling, structural data evaluation, and geotechnical core logging.
- **Rapid Geotechnical Assessments:** Standardized, rapid-deployment baseline assessments of ground conditions and preliminary hazard identification.
- **Practical Ground Control Recommendations:** Clear, concise, and implementable designs for underground development (support systems, extraction sequencing) and open-pit mining (slope stability, kinematic analysis).
- **Ground Control Frameworks & Risk Management:** We provide comprehensive support to develop ground control standards from the ground up, including customized training materials, advanced geotechnical risk assessments (such as Bowtie analysis and Fatal Risk Control Protocols), Codes of Practice (COP), and robust Ground Control Management Plans (GCMPs).
- **On-Site Personnel Training:** Hands-on workshops designed to empower mine workers and supervisors with fundamental rock mass characterization and hazard identification skills.
- **Tailings Storage Facility (TSF) Geotechnical Reviews:** Independent post-design assessments, seepage and drainage analysis, and advanced slope stability modeling under static and dynamic loading conditions.

Why Choose Apex Strata Consulting?

- **Technical Prowess:** We utilize advanced analytical tools to deliver highly accurate stability evaluations utilizing softwares including Rocscience suite, FLAC3D, and MAP3D amongst others.
- **Customer Focus:** Our deliverables are tailor-made to customer requirements, ranging from highly technical and complex reports to straightforward, heavily illustrated "what to do next" operational guides that your team can immediately implement.
- **Agile:** In an industry where conditions change rapidly, our streamlined operational model allows for rapid deployment to sites and vastly accelerated turnaround times on reports and recommendations.
- **Versatility:** From empirical underground designs to complex slope stability analysis, our expertise seamlessly spans both surface and underground environments.
- **Impartiality and Independence:** We provide unbiased, third-party reviews and QA/QC laboratory data reviews, ensuring regulatory compliance and genuine operational safety.
- **Experience:** Built on years of direct industry involvement, we deeply understand the

technical, financial, and operational constraints faced by small to mid-tier mining operators.

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Our People

Abercon Mbedzi – Principal Geotechnical Engineer



Abercon is a highly experienced geotechnical professional holding a Master's degree in Geotechnical Engineering from the University of Cape Town (UCT). With a sharp focus on advanced numerical modeling, slope stability analysis, and geotechnical domain characterization, Abercon leads Apex Strata's complex analytical projects. His expertise is central to conducting detailed slope stability analyses, designing effective ground support systems, and optimizing stopes sequencing for safe and efficient mining operations. Abercon applies a rigorous approach to kinematic and numerical analysis, ensuring that all designs are robust, practical, and aligned with industry safety standards and regulatory requirements.

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Freddy Chikwiri – Principal Geotechnical Engineer



Freddy is a highly experienced geotechnical professional holding a BSc (Hons) in Geology & Earth Science from the University of the Witwatersrand (South Africa) and a Master's degree in Mining Engineering (Rock Mechanics Specialization) from the prestigious Camborne School of Mines (UK). His experience ranges from large-scale PGM mines to small- and mid-tier tin and tungsten mines in East Africa. Freddy leads the integration and translation of advanced numerical modeling into actionable ground control designs, and specialized risk assessment frameworks, ensuring hazard-free operations that optimize both financial and operational efficiency.

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